Arab Academy for Science, Technology and Maritime Transport

College of Computing and Information Technology

Software Engineering Department

CSE4203 Principles of Software Architecture

Tamkeen

Inclusive Employment and 5% Quota Compliance Platform

Milestone 1: Proposal and Architectural Requirements

Team Name: Success Enablers

Submitted by:

Layal Ahmed Moussa	231006574
Mark Hany Elia	231006804
Sara Raef Abdelsalam	231015069
Youssef Sameh Elbendary	231006239
Zaid Mohamed Abdelrahman	231006511

Submitted to:

Dr. Nada Mahmoud Hassan

Eng. Mohamed Fareh

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1. System overview

The problem. Under Article 22 of Law No. 10 of 2018, any employer with 20 or more workers has to give at least 5% of its jobs to persons with disabilities (PwDs) [1], [3]. In practice the law is poorly enforced. Human Rights Watch reported this in June 2026, and by the end of 2025 only about 10% of PwDs (1.3 million people) had the Integrated Services Card that proves their status [3]. Some companies get around the quota with "paper jobs": they pay a PwD to stay at home so they can count them on the books [4]. Job seekers have their own problems. Most job sites don't work well with a screen reader, and many forms ask for a diagnosis.

Who uses it. Job seekers with visual, hearing, mobility or cognitive disabilities, sometimes with a job coach helping them. HR staff at companies. And verifiers on our side, who check that companies are real and that a person's disability card is valid.

What it does. Tamkeen is a job platform for PwDs, in Arabic and English. A job seeker's profile lists their skills and what they need to work well, such as a screen reader, step-free access or written instructions. We never ask for a diagnosis. Companies post jobs and say which accommodations they offer, and Tamkeen matches people to jobs that fit. When a company hires a verified PwD through the platform, its quota dashboard updates. A company with 120 employees, for example, needs 6 PwDs. If it has 2 verified, the dashboard shows a gap of 4.

Why it is architecturally interesting. On features alone, Tamkeen is a job board where people post, search and apply, and a single monolithic block could do that. What makes the architecture hard is a set of quality attributes that conflict with each other:

Trade-off	Where it shows up in Tamkeen
Security ⇄ Usability	The usual protections, like image CAPTCHAs, short session time-outs and extra login steps, lock out screen-reader users and users with cognitive disabilities. So we have to get security from things the user never sees, such as server-side authorization, rate limiting and audit trails.
Security (confidentiality) ⇄ Usability for employers	Law 151/2020 treats disability data as sensitive, so we need explicit consent that the user can withdraw [2]. Employers want details so they can plan accommodations, while candidates want to decide what is shared. That means the proof of disability status has to live apart from the profile employers see.
Availability ⇄ Cost	We run on free cloud tiers, and the free API server sleeps when nobody uses it. A job seeker still shouldn't lose an application because of that, and we can't pay for always-on backup servers.

Two other things shape the design. Many companies share one database, and none of them may see another's data. The quota rules come from the law and could change, so we want to update them without editing code.

2. Stakeholders and concerns

Each stakeholder cares about a different quality, which is why the scenarios in section 5 come from several of them. The last column names the quality attribute that matters most to each group.

Stakeholder	Who	Key concerns	Main quality attribute
PwD job seekers	Visual, hearing, mobility and cognitive disabilities	Works with their assistive technology; never asked for a diagnosis; control over who sees disability data; real jobs, not paper jobs	Usability, Security
Job coaches / family helpers	Parents, NGO job coaches	Can help without taking over; clear, revocable permission	Usability, Security
Employer HR recruiters	Recruiters, HR managers	Fast job posting; understanding accommodations; fast candidate search	Usability, Performance
Employer management	Owners, compliance officers	Correct quota figures; proof of compliance with Article 22; avoiding fines of 10,000 to 30,000 EGP and possible imprisonment [3]	Security (integrity)
Admins / verifiers	Platform team or a partner NGO	Fast verification queue; stopping fake companies; not keeping documents longer than needed	Security, Usability
Developers	The 5-person team	Simple stack, low coupling and high cohesion, easy to test, delivered on time	Modifiability, Testability
Operators	The team, as part-time operators	Free-tier limits, monitoring, easy redeploy and rollback	Availability
Disability regulator	National Council for Persons with Disabilities; Ministry of Social Solidarity	Quota computed per Law 10/2018; real employment; Integrated Services Card respected	Modifiability (rules), Security
Data-protection regulator	Personal Data Protection Center	Law 151/2020: consent, data minimization, security, transfers outside Egypt	Security
Wider Tamkeen platform	Graduation-project team and supervisor	Shared sign-in, clean API for the mobile app, consistent accessibility settings	Interoperability

3. System context (C4 Level 1)

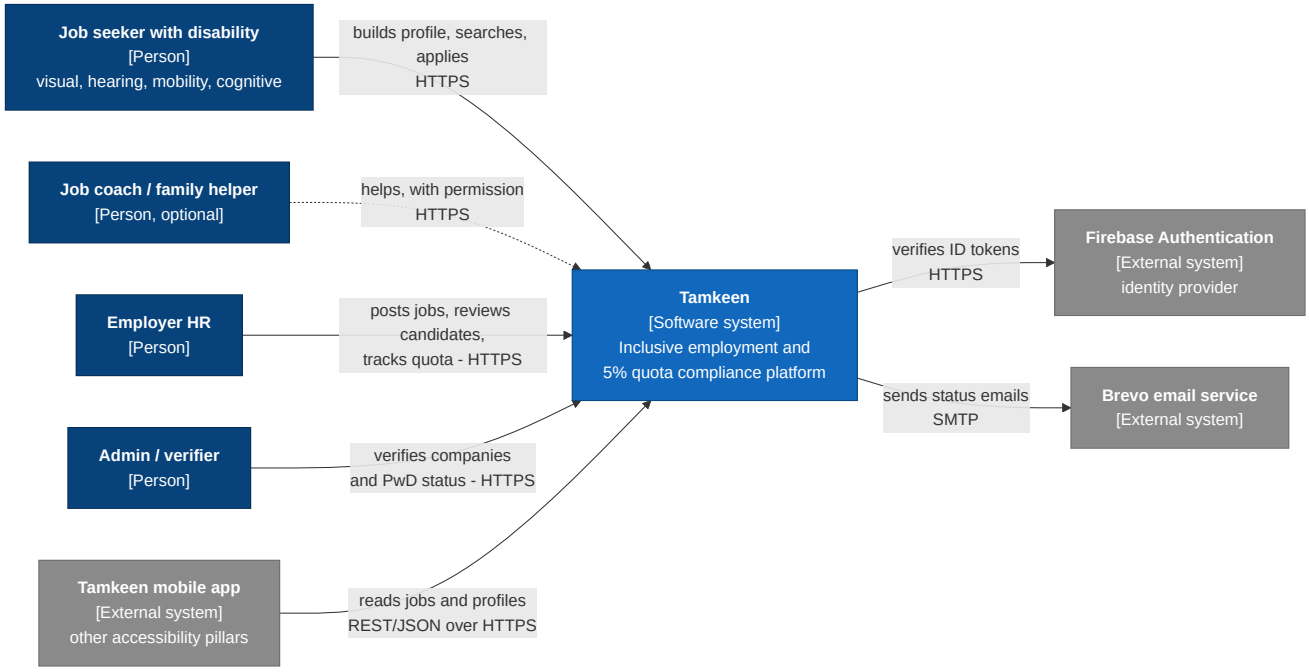


Figure 1. Tamkeen system context. Source file: diagrams/context.mmd

Person (user role) The system being built (one box) External system

Solid arrow: interaction, labelled with its purpose and protocol Dashed arrow: optional (Could priority)

External system	Role in the system	If it is unavailable
Firebase Authentication	Identity provider: one account across the Tamkeen platform; Tamkeen verifies its ID tokens	No new sign-ins; active sessions continue until tokens expire
Brevo email service (SMTP)	Application-status emails; the main contact channel for deaf users	Emails queue and retry; status stays visible in the app
Tamkeen mobile app	Shows jobs inside the wider platform through Tamkeen's REST API	No effect on the web app

4. Functional requirements

We wrote 15 user stories and ranked them with MoSCoW. In the M3 column, ✓ means we will build it in the Milestone 3 prototype, Stretch means we will build it if there is time, and × means it is not in the prototype. The Related QA column links each story to the quality attribute it affects most.

ID	User story	MoSCoW	M3	Related QA
FR-01	As any user, I sign in with my Tamkeen account and choose my role (job seeker, employer, verifier).	Must	✓	Security, Interoperability
FR-02	As a job seeker, I set language (Arabic/English), text size, contrast theme and easy-read mode before anything else.	Must	✓	Usability
FR-03	As a job seeker, I add skills, experience and work needs (e.g., screen reader, step-free access) without stating a diagnosis.	Must	✓	Usability, Security
FR-04	As a job seeker, I give explicit consent before disability-related data is stored, see who can view it, and withdraw or delete it at any time.	Must	✓	Security
FR-05	As an employer, I register my company with its headcount and post jobs that list the accommodations offered.	Must	✓	Usability
FR-06	As a job seeker, I search jobs by keyword, city and remote option, and filter by accommodations.	Must	✓	Performance
FR-07	As a job seeker, I apply to a job, choose which work needs to share with that employer, and track the status.	Must	✓	Availability, Security
FR-08	As an employer, I see a quota dashboard: required PwD headcount, current verified PwD staff, gap and percentage.	Must	✓	Security, Modifiability
FR-09	As a verifier, I approve or reject companies and PwD status (Integrated Services Card upload); the file is deleted after my decision.	Must	✓ (simplified)	Security
FR-10	As a job seeker, I see a match score with plain reasons, e.g., "offers screen-reader software; matches 4 of 5 skills".	Should	Stretch	Usability
FR-11	As a job seeker, I receive an email whenever my application status changes.	Should	Stretch	Interoperability
FR-12	As an employer, I export a monthly quota compliance report as PDF.	Should	×	Modifiability
FR-13	As a job seeker, I generate a screen-reader-friendly CV (HTML/PDF) from my profile.	Could	×	Usability
FR-14	As an employee, I anonymously review whether my job is real; companies receive a "reality score".	Could	×	Security
FR-15	As a job seeker, I connect with others, endorse skills and post to a social feed.	Won't	×	—

> The system has three main user roles: job seeker, employer HR and admin/verifier, plus an optional job coach. The prototype will cover all nine Must stories, so a user can go from signing in all the way to an updated quota dashboard.

5. Quality attribute scenarios

We wrote eight scenarios in the six-part form, covering six quality attributes: usability, security, availability, performance, modifiability and interoperability. We count accessibility under usability and privacy under security (confidentiality). Each response measure is something we can observe and test on the Milestone 3 prototype.

QAS-1 · Usability (accessibility): a blind user applies for a job

Source	Blind job seeker (end user) using the NVDA screen reader and a keyboard only
Stimulus	Wants to use the system efficiently: complete onboarding, build a profile and apply to a job
Artifact	Web user interface: the 6 core pages (sign-in, preferences, profile, search, job details, apply)
Environment	First use, normal operation, Arabic interface
Response	Every control is reachable by keyboard, labelled and announced in the correct language; feedback is spoken after each step; no CAPTCHA and no form time-out
Response measure	0 critical or serious axe-core violations on all 6 pages; Lighthouse accessibility score ≥ 95 per page; task completed without sighted help in ≤ 15 minutes in a walkthrough

Candidate tactics (Lecture 2): Support system initiative (provide feedback and assistance); support user initiative (pause / resume, cancel).

QAS-2 · Usability: a user with an intellectual disability applies in easy-read mode

Source	Job seeker with an intellectual disability (novice end user), helper nearby
Stimulus	Wants to find a job near home and apply
Artifact	Web user interface in easy-read mode
Environment	Normal operation
Response	One task per screen, short sentences with icons, progress saved after every step, no time limits
Response measure	Apply flow takes ≤ 6 screens; completed in ≤ 10 minutes in a walkthrough; ≥ 4 of 5 test users rate it "easy"

Candidate tactics (Lecture 2): Aggregate related actions; pause / resume; maintain a user model (easy-read preference stored once, applied everywhere).

QAS-3 · Security (confidentiality): an employer views an applicant

Source	Authenticated employer HR user (known user)
Stimulus	Attack to display information: tries to view an applicant's profile, including disability-related data
Artifact	Applications service and the sensitive-data store
Environment	Online, normal operation
Response	Returns only skills and the work needs the candidate chose to share for this application; disability status details, card files and consent records are never returned
Response measure	100% of employer-facing endpoints pass an automated test confirming sensitive fields are absent; 100% of stored sensitive records link to a consent record; after consent is withdrawn, data is hidden immediately and deleted within 24 hours (Law 151/2020)

Candidate tactics (Lecture 2): Resist attacks: authorize actors, limit exposure, encrypt data, separate entities; recover: maintain audit trail.

QAS-4 · Security (authorization): a company tries to read another company's data

Source	Authenticated HR user of Company A (malicious or faulty client)
Stimulus	Attack to access information: tries to read Company B's quota data or applicants by changing an ID in the request
Artifact	API authorization layer
Environment	Online, normal operation
Response	Request is denied (HTTP 403), no data is returned, and the attempt is logged with user ID and target
Response measure	100% of at least 20 cross-company test cases denied; 0 records from another company in any response; each denial logged within 1 second

Candidate tactics (Lecture 2): Resist attacks: authorize actors, limit access; detect attacks: detect intrusion; react: lock the account and inform actors after repeated attempts; recover: maintain audit trail.

QAS-5 · Availability: the server is asleep when someone applies

Source	Infrastructure (free-tier hosting platform)
Stimulus	Fault (omission): the API instance is asleep or restarting when a job seeker submits an application
Artifact	Web client and API process
Environment	Degraded mode: free-tier hosting, low traffic
Response	Fault is detected; the client keeps the draft on the device, shows an accessible "connecting, your answers are saved" message and retries; the server ignores duplicate submissions
Response measure	0 lost and 0 duplicated applications in 20 forced-restart tests; submission completes ≤ 90 seconds after the API wakes; monthly uptime ≥ 99% on a free uptime monitor

Candidate tactics (Lecture 2): Detect faults: ping/echo, heartbeat (uptime monitor); recover: retry, degradation; prevent: transactions (idempotent submit).

QAS-6 · Performance: job search under load

Source	Job seekers (external)
Stimulus	100 concurrent users submit filtered job searches (stochastic arrival)
Artifact	Job search service and database
Environment	Normal operation, free-tier hosting, API awake
Response	Matching jobs are returned, paginated, with the user's accessibility filters applied
Response measure	95% of requests answered in < 1.5 seconds; 0% failed requests during a 5-minute load test

Candidate tactics (Lecture 2): Control resource demand: limit event response (pagination), increase efficiency (database indexes); manage resources: maintain multiple copies of data (cache).

QAS-7 · Modifiability: the quota rule changes

Source	System administrator
Stimulus	Wishes to modify the quota rule (percentage, threshold or rounding) after a change in the law
Artifact	Quota engine rule configuration (data, not code)
Environment	Run time, production
Response	New rule version is added, applies from its effective date, past months keep their original results, with no side effects
Response measure	Change completed in ≤ 10 minutes; 0 code changes and 0 redeployments; 100% of quota unit tests pass for both rule versions

Candidate tactics (Lecture 2): Defer binding time: configuration files / runtime registration; increase cohesion: increase semantic coherence (one quota module owns all rule logic).

QAS-8 · Interoperability: the Tamkeen mobile app asks for job listings

Source	Tamkeen mobile app (a system)
Stimulus	Request to exchange information: asks for job listings and the user's application status, using a Firebase ID token
Artifact	Tamkeen REST API and the mobile app (the systems that interoperate)
Environment	Systems known before run time, normal operation
Response	Valid requests get JSON that matches the published OpenAPI contract; requests with a missing or expired token are rejected with HTTP 401 and logged
Response measure	100% of 50 contract-test requests correctly processed; 100% of 10 invalid-token requests correctly rejected and logged

Candidate tactics (Lecture 2): Manage interfaces: tailor the interface (one documented REST/JSON contract); detect and interpret errors from the other party (clear error codes).

6. Constraints

ID	Type	Constraint
C-T1	Technical	MERN stack (MongoDB, Express, React, Node.js); JavaScript throughout.
C-T2	Technical	Sign-in must use the Tamkeen platform's shared identity provider (Firebase Authentication).
C-T3	Technical	Must run on free tiers only: MongoDB Atlas free cluster, a free web-service host for the API (sleeps when idle), free static hosting for the web app, a free email tier.
C-T4	Technical	Must meet WCAG 2.1 Level AA [5] in Arabic (right-to-left) and English.
C-T5	Technical	Must expose a REST/JSON API, documented with OpenAPI, for the Tamkeen mobile app.
C-T6	Technical	No public government API exists to check the Integrated Services Card, so verification is manual.
C-B1	Business / legal	Law 10/2018, Article 22: employers with ≥ 20 workers must employ ≥ 5% PwDs [1], [3]. The law does not specify rounding; we round up (30 workers → 2 required).
C-B2	Business / legal	Law 151/2020: disability data is sensitive personal data; explicit consent, right to withdraw, limits on transfer outside Egypt [2]. Free cloud tiers are hosted abroad, so the prototype uses synthetic data only.
C-B3	Business	No payments or fees in this phase.
C-B4	Business	Never ask for a diagnosis; collect work needs instead.
C-O1	Organizational	Five part-time students with beginner-to-intermediate MERN skills; no dedicated DevOps member.
C-O2	Organizational	Deadlines: M1 on 23 October 2026; project discussion in university weeks 12 and 13 (5 to 18 December 2026).
C-O3	Organizational	All work submitted through GitHub; documents as PDF.
C-O4	Organizational	The system is also the employment pillar of the team's Tamkeen graduation project and must fit its shared sign-in and settings.

7. Architecturally significant requirements

We used the rule from Lecture 1: a requirement is architecturally significant when a quality attribute drives it and the decision behind it would be expensive to change later. Each scenario in the tree is rated (Importance, Difficulty). Importance is how much it matters to the customer (business value). Difficulty is how hard it is for us to achieve (architectural impact).

Utility tree

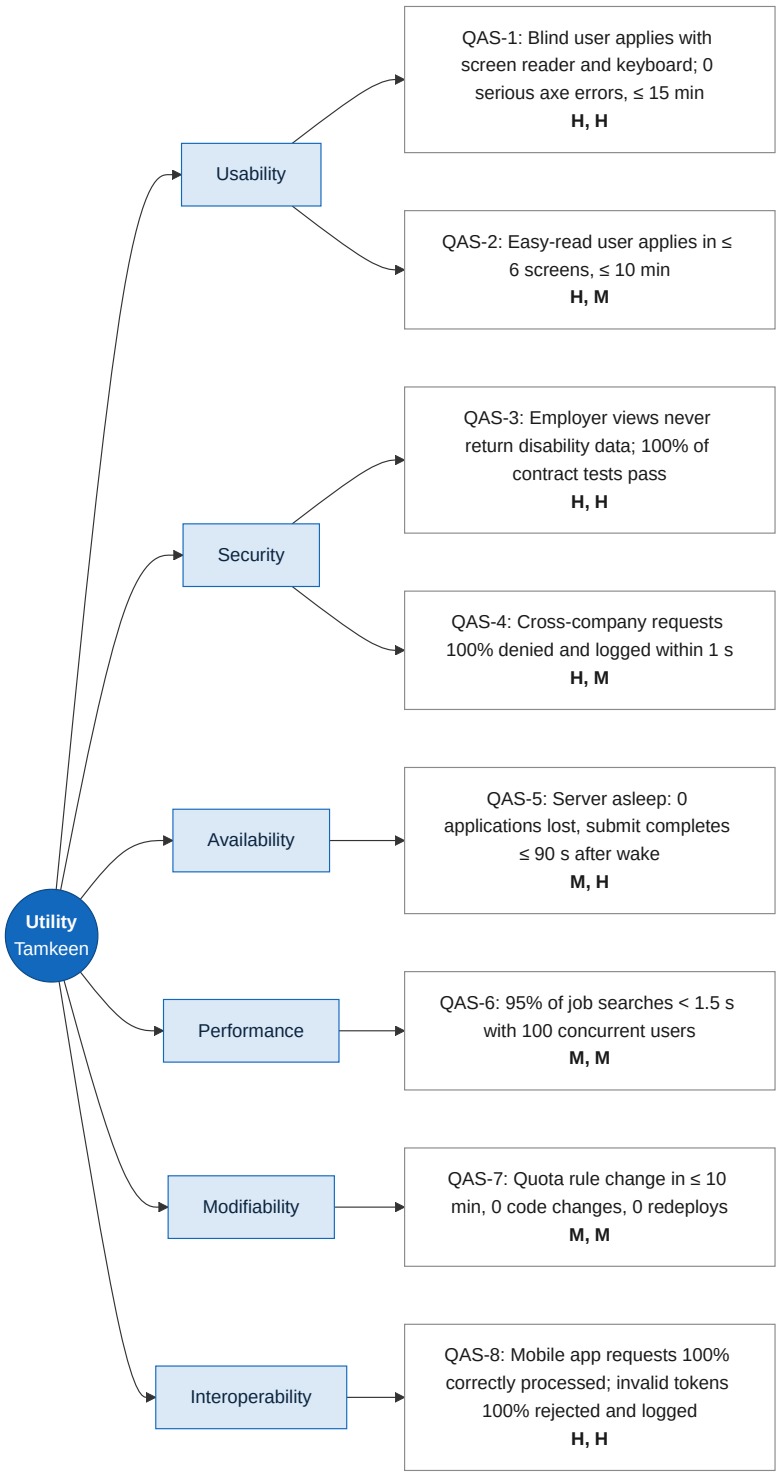


Figure 2. Utility tree for Tamkeen: quality attribute → scenario → (Importance, Difficulty). Source file: diagrams/utility-tree.mmd

Prioritized requirements (sections 4 to 6)

Rank	Requirement	Quality attribute	Business value	Arch. impact	Why it is architecturally significant
1	QAS-3, FR-04, C-B2: confidentiality of disability data	Security	**H**	**H**	Needs a separate, encrypted store for sensitive data, consent records, and different API responses per role. Adding this after the data model exists would mean changing every module that reads profiles.
2	QAS-1, QAS-2, FR-02, C-T4: accessible use for all disability types	Usability	**H**	**H**	Needs one shared library of accessible UI components, Arabic and English (with RTL) handled once at the app shell, and user preferences stored in one place. Fixing accessibility screen by screen later costs much more.
3	QAS-8, FR-01, C-T2, C-T5: shared Tamkeen identity, API-first	Interoperability, Security	**H**	**H**	Needs an external identity provider, a token check on every request, users identified by an external ID, and a documented API that the mobile app can also call.
4	QAS-4, FR-08: company isolation	Security	**H**	**M**	Needs one authorization layer that limits every query to the user's own company. Checks spread across handlers are easy to forget.
5	QAS-5, C-T3: availability on free hosting	Availability	**M**	**H**	Needs drafts saved on the client, automatic retry, and submit endpoints that are safe to call twice, since we can't keep the server awake.
6	QAS-6, FR-06: search response time	Performance	**M**	**M**	Needs indexes, pagination and caching; can be tuned later without restructuring.
7	QAS-7, C-B1: configurable quota rules	Modifiability	**M**	**M**	Rules are stored as versioned data in one quota module.
8	FR-09, C-T6: manual verification	Security	**H**	**M**	Needs a verification queue, short-lived file storage and an audit trail.
9	FR-10: explainable matching	Usability	**M**	**L**	Self-contained and fits in any reasonable structure.
10	FR-11, FR-12: email, PDF report	Interoperability	**M**	**L**	Can sit behind an interface and be added later.

Ranks 1 to 5 are our ASRs, and they will drive the design in Milestone 2. We still have to meet ranks 6 to 10, but they will fit inside whatever structure the top five lead to.

Contribution table

Member	Contribution	Reviewed
Zaid Mohamed Abdelrahman	Section 1 System overview; Section 7 ASRs and utility tree; final editing	Sections 4, 5
Layal Ahmed Moussa	Section 2 Stakeholders and concerns; references	Sections 1, 3
Mark Hany Elia	Section 3 System context diagram and diagram sources; Section 6 Constraints	Sections 2, 7
Sara Raef Abdelsalam	Section 4 Functional requirements; QAS-1, QAS-2 (usability)	Section 6
Youssef Sameh Elbendary	Section 5: QAS-3 to QAS-8 (security, availability, performance, modifiability, interoperability)	Section 4

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